

powerful. Consider a general R transformation

$$\theta_L \rightarrow e^{i\varphi} \theta_L, \quad \Phi \rightarrow \Phi, \quad V_A \rightarrow V_A, \quad Y \rightarrow e^{2i\varphi} Y, \quad (29.3.6)$$

with arbitrary real φ . This leaves the T -independent terms of the Lagrangian density (29.3.3) invariant, but according to Eq. (29.3.4), quantum effects violate this symmetry, just as if there were a term $\Delta\mathcal{L}$ in the Lagrangian density with a transformation

$$\Delta\mathcal{L} \rightarrow \Delta\mathcal{L} - \frac{1}{32\pi^2} (C_1 - C_2) \sum_A \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_A^{\rho\sigma} \varphi.$$

Recalling Eq. (27.3.18), this is cancelled if we give T a transformation

$$T \rightarrow T + (C_1 - C_2)\varphi/\pi. \quad (29.3.7)$$

Because $W_{A\alpha L}$ has $R = 2$, the whole theory including non-perturbative effects is invariant under the *combined* transformations (29.3.6) and (29.3.7). In particular, the superfield $\exp(2i\pi T)$, which for $T = \tau$ is periodic in θ , has $R = 2(C_1 - C_2)$.

We again introduce an ultraviolet cut-off, and consider the effective 'Wilsonian' Lagrangian

$$\begin{aligned} \mathcal{L}_\lambda^\# = & \left[\mathcal{A}_\lambda(\Phi, \Phi^\dagger, V, T, T^\dagger, Y, Y^\dagger, \mathcal{D} \cdots) \right]_D \\ & + 2 \operatorname{Re} \left[\frac{T}{8\pi i} \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} + \mathcal{B}_\lambda(\Phi, W_L, T, Y) \right]_{\mathcal{F}}, \end{aligned} \quad (29.3.8)$$

with $\mathcal{A}_\lambda$ and $\mathcal{B}_\lambda$ both gauge-invariant functions of the displayed arguments. The term proportional to T has been separated from the function $\mathcal{B}_\lambda$ in order that the translation (29.3.7) of T should continue to cancel the anomaly in the R transformation (29.3.6). Invariance under the combined transformation (29.3.6), (29.3.7) then tells us that terms in the function $\mathcal{B}_\lambda$ must be proportional to powers of $\exp(2i\pi T)$, which have definite R values.

Furthermore, it is only *positive* powers of $\exp(2i\pi T)$ that may appear in $\mathcal{B}_\lambda$. According to Eq. (27.3.24), it is only instantons with positive winding number $\nu \geq 0$ that can make contributions to the effective Lagrangian that are holomorphic in T rather than T^* , and these give rise to factors $\exp(2i\pi\nu T)$. More generally, for $T = \tau$ any power $\exp(2i\pi a T)$ will depend on the gauge coupling through a factor $\exp(-8\pi^2 a/g^2)$, so that a must be positive in order for non-perturbative effects to be suppressed for small g . In consequence, non-perturbative effects enter in $\mathcal{L}_\lambda^\#$ through operators $\exp(2i\pi a T)$ that have positive-definite, zero, or negative-definite values of R , depending on whether $C_1 > C_2$, $C_1 = C_2$, or $C_1 < C_2$. (In the generalized supersymmetric version of quantum chromodynamics